Chapter 2 Na Beans(S1-E1)

1. Write a step-by-step algorithm (in plain English, no code) for making a cup of tea. Your algorithm must include at least one decision step (if/else) and one repetition step (loop).

2. A student must cross a road safely. Write a precise algorithm with numbered steps that includes: checking for traffic, waiting if needed, and crossing when safe. Identify which step is a loop and which is a conditional.

3. Translate this everyday instruction into pseudocode: If it is raining, take an umbrella; otherwise, put on sunscreen.

4. Consider an ATM machine. Without writing any code, describe the algorithm for withdrawing cash. What data does it need as input? What decisions does it make? What are the possible outputs?

5. Write a Python expression that correctly computes the average of five numbers: 10, 20, 30, 40, 50. Many beginners write (10 + 20 + 30 + 40 + 50 / 5) – identify the bug and fix it.

6. Explain string concatenation with +. Write a script where the user inputs their first name and last name separately. Combine them and print a greeting like: Welcome, Adeola Grace!

7. Write a Python program that takes a mark (0–100) as input and prints the corresponding letter grade: A (90–100), B (80–89), C (70–79), D (60–69), F (below 60).

8. Write a Python script to calculate simple interest. Inputs: principal, rate (%), and time (years).
Formula: SI = (P x R x T) / 100. Also display the total amount A = P + SI.

9. Write a Python program that asks the user for a number of seconds and converts it into hours, minutes, and remaining seconds. Example: 3661 seconds = 1 hour, 1 minute, 1 second.

10. Draw or describe using words a flowchart that determines whether a student passes or fails an exam. The pass mark is 50. Include: Start, Input mark, Decision, two output paths, and End.